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Advancing Natural Fiber-Reinforced Composites Through Incorporating ZnO Nanofillers in the Polymeric Matrix: A Review

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ABSTRACT

Natural fiber-reinforced composites have gained significant popularity owing to their lightweight nature, low environmental impact, favorable specific properties, and cost-effectiveness. However, these composites often lack sufficient mechanical strength, thermal resistance, water repellence, and antibacterial performance, limiting their applicability in a variety of fields. To address these limitations, researchers have recently investigated the incorporation of ZnO nanofillers into the polymeric matrix of natural-fiber-reinforced composites. Nevertheless, there is a gap in the current literature regarding a review on the utilization of ZnO nanoparticles as fillers in matrix materials to improve the performance of natural-fiber-reinforced composites. Therefore, this review discusses recent advancements in incorporating ZnO nanofillers into natural fiber composites, with an emphasis on how these advancements affect the mechanical, water, thermal, and antibacterial properties of the composites. This review explores the methods for incorporating ZnO nanofillers into a matrix, including solution and melt mixing, and discusses the influence of surface modification on mitigating agglomeration issues. The potential applications and future perspectives of ZnO nanofiller-reinforced natural fiber composites in the automotive, construction, and packaging industries are also discussed.

摘要

天然纤维增强复合材料因其轻质、低环境影响、良好的比性能和成本效益而广受欢迎。然而，这些复合材料往往缺乏足够的机械强度、耐热性、防水性和抗菌性能，限制了其在各种领域的适用性。为了解决这些局限性，研究人员最近研究了将ZnO纳米填料掺入天然纤维增强复合材料的聚合物基质中。然而，在目前的文献中，关于利用ZnO纳米颗粒作为基质材料中的填料来提高天然纤维增强复合材料的性能的综合述存在空白。因此，本综述讨论了将ZnO纳米填料掺入天然纤维复合材料的最新进展，重点讨论了这些进展如何影响复合材料的机械、水、热和抗菌性能。这篇综述探讨了将ZnO纳米填料掺入基质的方法，包括溶液和熔体混合，并讨论了表面改性对减轻团聚问题的影响。还讨论了ZnO纳米填料增强天然纤维复合材料在汽车、建筑和包装行业的潜在应用和未来前景。

KEYWORDS

Natural fiber reinforced composites; ZnO nanoparticles; matrix filler; mechanical properties; thermal properties; water repellency; antimicrobial properties

关键词

天然纤维增强复合材料; ZnO纳米粒子; 基质填料; 机械性能; 热性能; 防水性; 抗菌性能

Introduction

Material scientists are focusing on creating polymer matrix composites with natural or synthetic fibers and fillers for their strength, durability, and lightweight nature in various applications (Ramesh et al. 2022). Notably, in recent years, there has been a surge in the development of environmentally friendly composite materials, with natural fibers being a popular choice owing to their low weight, cost, improved mechanical properties, and biodegradability (Dejene and Geletaw 2024; Mishra et al.

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2022). Despite the various benefits associated with natural fiber-reinforced composites, they have certain drawbacks including inadequate interfacial adhesion, limited moisture resistance, dimensional and thermal instability, poor durability, and poor mechanical properties (Amjad et al. 2022; Dejene 2024a). Consequently, these inherent drawbacks impede the widespread use of natural fiber-reinforced composites in high-performance and enduring applications (Akter, Uddin, and Anik 2024).

Hence, researchers have developed multiple methods to improve the properties of natural-fiber-reinforced composites for diverse applications. One approach involves reinforcement of polymers with organic and inorganic fillers to enhance the toughness of the matrix material (Bag et al. 2024; Ramesh et al. 2022; Rashid et al. 2024). Another method focuses on chemically modifying fibers to improve the fiber-matrix interface using alkali treatment, coupling agents, and nanoparticles in surface treatment (Dejene 2024a; Mohammed, Rahman, Mohammed, Adam, et al. 2022). Among these methods, the incorporation of nanofillers through advanced manufacturing techniques has significantly improved natural fiber-reinforced composite materials by modifying the interaction between the fiber and matrix (Adib et al. 2024; Siddiqui, Rabbi, and Rahman 2024).

Nanofillers have the potential to improve the mechanical and functional performances of composites, thereby expanding their applications in various fields. In the realm of nano-based natural fiber-reinforced composites, a larger aspect ratio of nanofillers provides better reinforcement, leading researchers to become more interested in this material (Hasan, Horváth, and Alpár 2020). These nanofillers are inorganic or organic materials that are used in the production of composites. They can be inorganic, such as alumina, magnesia, silica, zinc oxide, titanium dioxide, and calcium carbonate, or naturally organic, such as carbon black and cellulosic fibers. The incorporation of nanofillers results in the reduction of available free spaces, increases the stiffness of the laminates, and facilitates bonding between the matrix and fibers, thereby promoting improved interaction (Amjad et al. 2022). Natural fiber-reinforced composites (NFRCs) containing nanoparticles (NPs) are characterized by their eco-friendly nature, reduced water absorption, and enhanced mechanical properties. They are used in the construction, transportation, aerospace, and other consumer products. Furthermore, incorporating specific nanomaterials into NFRCs can impart additional functionalities such as antibacterial, anti-odor, UV protection, and highly hydrophobic properties (Mishra et al. 2022).

Zinc oxide nanoparticles (ZnO NPs), along with other types of nanoparticles, play a significant role in enhancing the properties of the composites. The incorporation of ZnO NPs in composites has been found to improve composite properties, including barrier, mechanical, and antimicrobial activities. They have been widely studied for their versatile applications, offering advantages such as antibacterial activity, piezoelectricity, corrosion resistance, thermal conductivity, and their potential as structural supercapacitors (Lorero et al. 2020). Despite the existing body of literature on composite materials, a notable gap remains in the literature concerning the effects of incorporating ZnO nanofillers into NF-based composites. Given the growing demand for natural-fiber-reinforced composites with enhanced properties, this gap represents a crucial area for further investigation. Therefore, the author is motivated to address this research gap and to present a comprehensive review based on significant findings in the field.

Review methodology

Researchers and material scientists are currently focusing on creating advanced materials that satisfy the requirements of robustness, manufacturability, affordability, and lightweight. Research is ongoing on the potential application of natural fiber-reinforced polymer matrix composites in engineering and technology owing to their enhanced performance compared to conventional materials. Accordingly, both industry and academia have focused on incorporating organic and inorganic nanofillers into composites to improve their properties and potential applications. Nevertheless, there is a gap in the current literature regarding the utilization of ZnO NPs as fillers in matrix materials to improve the performance of natural-fiber-reinforced composites. Therefore, this review is solely dedicated to analyzing the impact of ZnO nanofillers on composite properties and considering the potential applications of ZnO nanofiller-based natural-fiber-

reinforced composites. This review encompasses a thorough examination of both original research and review articles that focus on composites composed of natural fibers, ZnO nanofillers, and polymer matrices. This review followed an iterative search process that updated search terms as the review progressed. The literature selection was based on peer-reviewed scientific articles, reviews, book chapters, and conference proceedings. Relevant articles were selected for full-text assessment and thoroughly reviewed and analyzed to extract key information. Thus, providing readers with a comprehensive review aims to address key issues in the literature by providing an organized and thorough analysis of the findings of the most significant studies. This review is structured to provide a concise overview of NFRCs, outlining their limitations and the ongoing efforts within the scientific community to overcome these challenges. Subsequently, the review explored the properties, surface modification and techniques for incorporating ZnO nanofillers into the polymer matrices. It then delves into the properties of ZnO nanofiller-based plant-fiber-reinforced composites, focusing on their mechanical, thermal, water absorption, and antimicrobial properties. Finally, this review summarizes the outcomes of the reviewed studies and proposes avenues for future research.

Natural fibers in composite materials: properties and issues

Composition and properties of natural fiber

Natural fibers are extracted from various renewable resources, including animals, vegetable plants, and minerals (Mochane et al. 2019). From a composite standpoint, plant fibers, such as basts, leaves, and fruits exhibit the most desirable properties for biocomposites (Akter, Uddin, and Anik 2024). Plant fibers typically include cellulose (60–80%), hemicellulose (20–30%), and lignin (5–20%), with the remainder consisting of wax, pectin, moisture, and water-soluble organic components (Balla et al. 2019). The chemical composition of natural fibers, including their crystallinity, microfibrillar angle, defects, and physical properties, significantly influence their performance. These fibers are susceptible to biological, chemical, mechanical, thermal, photochemical, and aqueous degradation, depending on their constituents (Mochane et al. 2019).

Physical properties of plant fibers

Plant fibers, including leaves, fruits, stems, grasses, and wood, have physical properties, such as dimensions, density, moisture absorption, and physical structure. They are suitable as fillers in engineering and non-engineering applications because of their lightweight nature, compatibility, porous structure, and affordability. However, as the fiber type varied, the cellulose content and crystallinity also changed, as shown in (Table 1) (Balla et al. 2019).

Mechanical properties of plant fiber

The natural fiber structure and dimensions, including the density and microfibril angle, directly affect the mechanical properties of NFRCs, which depend on the inherent properties of natural fibers (Li et al. 2020). Figure 1a illustrates that high-density fibers frequently exhibit higher strength and stiffness than low-density fibers do. Similarly, the elastic modulus of natural fibers strongly depends on the microfibrillar angle (MFA), with a low MFA resulting in stiffer fibers (Figure 1 (b,c)). This is because a low MFA enables cellulose fibrils to be almost parallel to the loading axis and can support more load, thereby increasing the stiffness. However, owing to their high stiffness, these low-MFA fibers typically exhibit brittle behavior. However, natural fibers with a high MFA typically exhibit a large plastic deformation and high toughness (Balla et al. 2019). The diameter of the fibers and the level of cellulose polymerization are additional factors that affect the mechanical characteristics of the natural fibers. (Table 1) summarizes the chemical constituents and physical and mechanical properties of plant-based natural fibers (Dejene and Geletaw 2023a; Yadav, Dixit, and Dixit 2023).

Table 1. Chemical constituents and physical and mechanical properties of the plant-based natural fibers (Dejene 2024a, 2024b.)

Fibers	Composition			Physical properties		Mechanical properties		
	Cellulose (wt. %)	Hemicellulose (wt. %)	Lignin (wt. %)	Moisture content (wt. %)	Density (g/ cm ³)	Tensile strength (MPa)	Young's modules (GPa)	Elongation at break (%)
Flax	64.1	16.7	2.0	8–12	1.4	800–1500	60–80	1.2–1.6
Ramie	68.6	13.1	0.6	7.5–17	1.5	500	44	2
Hemp	74.4	17.9	3.7	6.2–12	1.48	550–900	70	1.6
Jute	64.4	12	11.8	12.5–13.7	1.46	400–800	10–30	1.8
Coir	32–43	0.15–0.25	40–45	8	1.25	220	6	15–25
Sisal	65.8	12	9.9	10–22	1.33	600–700	38	2–3
Banana	55.2	16.8	25.3	20–28	1.2	20–290	19.7–27.1	1.1
Henequen	60	28	8	10–12	1.4	430–580	–	3–4.7
Pineapple	81	–	12.7	14	1.5	170–1627	82	1–3
Banana	60–65	19	5–10	8–10	1.35	355	33.8	5.3
Kenaf	53.4	33.9	21.2	6.2–12	1.2	295	–	2.7–6.9
Cotton	82.7	5.7	–	7.85–8.5	1.5	400	12	3–10
Bamboo	48.2–73.8	12.5–73.3	10.2–21.4	11–17	1.1	500–575	27–40	1.9–3.2
Enset	56.37–70.5	22.47–27.88	4.8–12.3	7.8–13.84	1.099–1.61	240–900	12–69	0.48–3.24

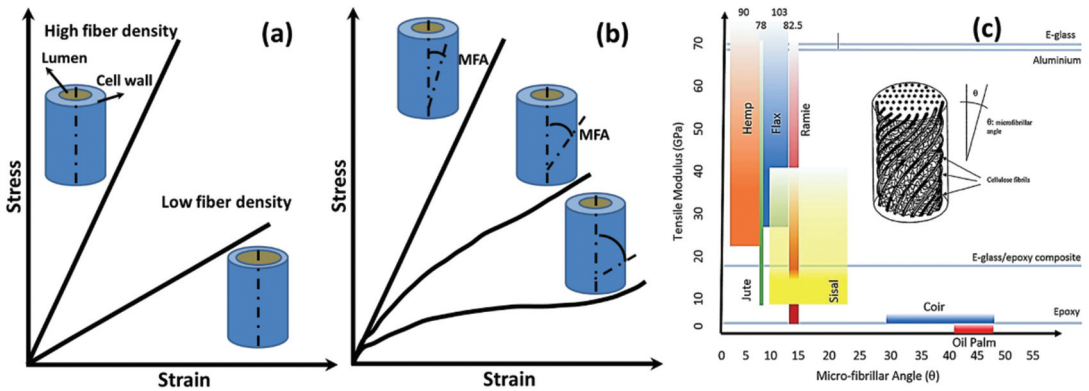


Figure 1. Influence of fiber density (a) and microfibril angle (MFA) (b) on stress-strain behavior and mechanical properties of natural fibers (c) Tensile modulus of different natural fibers as a function of their microfibril angle (Balla et al. 2019). Reprinted with the permission from Elsevier.

Polymer/matrix materials

A matrix connects the reinforcing components in a composite, and plant fibers are combined with a polymeric matrix to create eco-friendly composites. The matrix holds the fibers in place and prevents cracks and damage (Akter, Uddin, and Anik 2024). For polymeric matrices, thermosets or thermoplastics may be used because of their advantageous properties such as low density, low electrical and thermal conductivity, and good corrosion resistance. However, fillers have also been added to improve the performance (mechanical characteristics) of polymers and reduce the cost of materials (Nayak, Kumar Nayak, and Panigrahi 2021). (Table 2) shows the properties of most commonly used polymeric matrices.

Thermoplastics, such as polyethylene, polypropylene, and polyvinyl chloride, are commonly utilized as matrices for natural fiber composites because of their low melting temperatures, whereas thermosets, including phenolic, polyester, and epoxy resins, are most frequently used in NFRCs (Yousry et al. 2021). When heated, thermoplastics soften and can be remolded without significant degradation into a variety of products. Therefore, recyclability was the most important characteristic. However, thermoset matrices cannot be recycled or their shape cannot be changed once polymerization (curing) is

Table 2. Main characteristics of thermosetting and thermoplastic matrix (Mishra et al. 2022.)

Polymers	Density(g/cm ³)	Tensile modulus (GPa)	Tensile strength (MPa)	Compression strength (MPa)
Thermosetting				
Polyester	1.0–1.5	2.0–4.5	40–90	90–250
Epoxy	1.1–1.6	3.0–6.0	28–100	100–200
Vinyl ester	1.2–2.4	3.1–3.8	69–86	86
Phenolic	1.29	2.8–4.8	35–62	210–360
Thermoplastic				
				Melting temperature (°C)
Polypropylene (PP)	0.90–0.91	1.1–1.6	20–40	175
Polyethylene (PE)	0.91–0.95	0.3–0.5	25–45	115
Polyvinylchloride (PVC)	1.38	3.0	53	212
Polystyrene (PS)	1.04–1.05	2.5–3.5	35–60	240
High-density PP	0.94–0.97	0.5–1.1	30–40	137
Recycled PET	1.38–1.45	2.4–4.1	50–100	245–255

completed. The main thermoset resins used for natural fiber composites are epoxy and unsaturated polyesters because of their good interactions and better wettability with natural fibers, which attain good strength during the application of the developed composite material (Neto et al. 2021; Radhakrishnan et al. 2023; Yousry et al. 2021). Epoxy resin, a polymeric matrix, offers chemical, corrosive, and strong adhesive properties. However, they exhibit brittleness, shrinkage, and low-impact properties, making them unsuitable for high-performance aircrafts and automobiles. The demand for particulate fillers and polymer materials for industrial and structural applications has increased (Ashok, Kalaichelvan, and Damodaran 2022). Therefore, researchers are exploring nanosized fillers such as ZnO NPs, carbon black, clay, and nanofibers to expand their use in high-tech fields (Dhanapal et al. 2023). Nanocomposites of polymer matrices reinforced with semiconductor nanoparticles are gaining interest because of their improved optical and electrical properties. Zinc oxide nanostructures are becoming popular in scientific and industrial fields because of their physical properties, particularly their electrical, optical, and piezoelectric properties (Matysiak, Tański, and Zaborowska 2019).

Natural fiber reinforced polymeric composites (NFRPCs)

Fiber-reinforced polymer composites (FRPC) are lightweight, strong materials with enhanced mechanical properties (Dhanapal et al. 2023). NFRCs are gaining popularity owing to their low environmental impact and low cost. Plant fibers are a promising research topic because of their low price, low carbon emissions, abundant availability, low density, bio-renewable traits, and low energy consumption. They are used in construction, aerospace, ballistics, wind energy, and automotive (Akter, Uddin, and Anik 2024; Dejene 2024a, 2024b). However, natural fibers have limitations, such as low thermal stability, high flammability, high moisture absorption, and variation in mechanical properties. To overcome these limitations, studies have focused on combining natural fibers with other fillers such as nanofillers. The choice of nanofiller depends on the intended application to overcome the shortcomings of natural fibers (Mochane et al. 2019). Akter et al. (2024) highlighted the growing trend of polymer composites reinforced with natural fibers (according to the Web of Science, accessed in 2022) using keywords such as natural fibers, plant fibers, reinforced polymers, and composites, as shown in Figure 2.

Strategies to enhance NFRPCs properties

Natural fibers, which are hydrophilic, have poor interfacial interactions with hydrophobic polymeric materials, limiting the stress transfer between composite components. Chemical and physical modifications have been explored to improve interfacial adhesion and overall composite properties. The treatment of natural fibers can improve the biodegradation stability and mechanical performance (Mochane et al. 2019). However, these methods do not always yield the desired performance, thermal stability, or



Figure 2. The number of NFRCs-related publications and citations in the last 10 years (Akter, Uddin, and Anik 2024).

barrier resistance. Incorporating nanofillers into NFRCs can eliminate these drawbacks. Combining organic natural fibers with inorganic or organic polymers and nanoparticles has the potential to improve the mechanical performance and expand applications (Hasan, Horváth, and Alpár 2020). Researchers have found that adding nanofillers to a composite matrix enhances fiber-matrix interfacial interactions, improves matrix toughening properties, and reduces weight simultaneously (Baghdadi et al. 2020; Nayak, Kumar Nayak, and Panigrahi 2021).

Nano-based natural fiber reinforced polymeric composites

Composites often contain natural or artificial fillers in various forms such as particles, fragments, fibers, sheets, and whiskers (Mishra et al. 2022). Nanoparticles or nanofillers have the potential to improve the thermal and mechanical properties of polymer composites, flame retardancy, and reduce moisture adsorption (Mohammed, Rahman, Mohammed, Adam, et al. 2022). They also offer exclusive flexible functionalities and better mechanical strengths than pure materials (Kausar 2021). Researchers are increasingly using nanomaterials in composite preparation owing to their large relative surface area and quantum effect, which enhance material properties, such as chemical reactivity, heat resistance, strength, and optical, electrical, and magnetic behaviors. Nanomaterials can exist in single- or multi-stage forms, with single-phase materials typically having at least one of the dimensions smaller than 100 nm. Multiphase nanocomposites can be engineered by adding nanomaterials to the matrix, forming the basis of these composites (Mishra et al. 2022; Serkan 2019; Sridhar, Gobinath, and Kirgiz 2022).

Nanoparticle-embedded natural-fiber-reinforced composites perform well at high temperatures without altering the processing conditions or melting temperatures. Some thermoset polymers become fragile owing to crystallization; however, this issue can be eliminated by adding biofibers and nanofillers (nano TiO_2 , SiO_2 , carbon nanotubes, ZnO , and graphene oxides). Adding a nanofiller to the polymer matrix enhances the composite density and hardness of NFRCs. Natural fibers have better specific properties and are lighter than synthetic fibers, which, when combined with another reinforcing agent (nanofiller), enhances the performance of nano-biocomposites (Hasan, Horváth, and Alpár 2020). Various nanoparticle applications are currently being investigated for their use as fillers in natural fiber-reinforced polymer composites. ZnO NPs are an interesting candidate due to

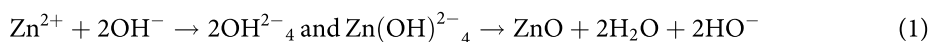
their large surface area, non-toxicity, availability, low cost, stability, high ultraviolet absorption capacity, and strong antimicrobial activity (Shankar, Wang, and Rhim 2018). This review is necessary because of the extensive use of ZnO NPs fillers in NFRCs by numerous researchers.

Why ZnO nanoparticles?

Metal-oxide nanocomposites are of special interest because of their distinctive thermal, mechanical, optical, magnetic, electronic, and catalytic properties. Zinc oxide (ZnO), one of the many different types of metal oxide semiconductors, has attracted interest because of its exceptional qualities, such as high optical and thermal stability, low toxicity, excellent physical and chemical stability, and widespread availability (Quadri et al. 2017; Zongyu, Bockstaller, and Matyjaszewski 2021). ZnO is an inorganic substance that is insoluble in water and is typically found as a white powder. It is used as an additive in a variety of substances and goods such as lubricants, cosmetics, plastics, rubbers, ointments, food supplements, paper, pigments, batteries, ceramics, fire retardants, first-aid tapes, and cement (Zongyu, Bockstaller, and Matyjaszewski 2021).

ZnO is a binary compound with a wide bandgap (3.37 eV) and wurtzite structure. Because of their piezoelectric properties, they have potential applications in sensors and micromechanical systems. ZnO NPs have also been extensively studied for their applications in biomaterials, medicine, wastewater treatment, and electronics (Zongyu, Bockstaller, and Matyjaszewski 2021). A study on ZnO nanomaterial biocompatibility, antimicrobial performance, and multiphase morphology of biomimetic nanocomposites materials with ZnO/sodium alginate/hydroxyapatite-oriented granules and ZnO/hydroxyapatite hydrogels demonstrated the Zn²⁺ behavior of different composite materials (Turlybekuly et al. 2019).

The formation reaction of ZnO was described by the following authors:



In a previous study, the average particle size of ZnO ranged from 12 nm to 30 nm. The distribution of the nanoparticle was homogeneous, along with a very good compatibility in terms of the thermal property (Hasan, Horváth, and Alpár 2020).

Synthesis of ZnO nanoparticles

Numerous synthetic methodologies for the preparation of ZnO with different nanostructures have been demonstrated, including “dry” methods and “wet” solution-phase methods. Examples of synthetic methods include the sol-gel technique, spray pyrolysis, organic precursors, thermal decomposition, plasma synthesis, hydrothermal processing, vapor transport process, microwave-assisted syntheses, direct precipitation, and homogeneous precipitation. Wet chemical methods enable control of the size and shape of ZnO nanocrystals by varying the solvent, reactant concentration, and temperature (Zongyu, Bockstaller, and Matyjaszewski 2021).

Torres et al. (2022) obtained ZnO NPs with wurtzite morphology and particle sizes between 50–100 nm using a coprecipitation method. ZnO NPs were incorporated to the biolaminates of Ixtle and Henequen natural fibers with bio-based epoxy resins to evaluate their mechanical properties and fracture behavior. Dhanapal et al. (2023) synthesized nanosized ZnO using homogeneous precipitation and calcination methods to study the effects of amine-functionalized nano-ZnO-reinforced areca fiber/epoxy hybrid nanocomposites. Abdullah et al. (2017) reported that the sol-gel method is suitable for synthesizing nanomaterials with a size of less than 50 nm. Devaraju and Sivasamy (2018) utilized the sol-gel method to create nano-based sisal fiber-reinforced composites using ZnO NPs and zirconium oxide nanoparticles (ZrO₂-NPs). Similarly, the sol-gel method has been used to synthesize ZnO NPs for the production of nano-based palm fiber-reinforced epoxy composites (Devaraju, Sivasamy, and Loganathan 2020). The use of alternative recycling synthesis to recover materials such as ceramic oxides opens an interesting opportunity to create composites in more sustainable

ways. Lorero et al. (2020) obtained a ZnO nanofiller from spent alkaline batteries using a new recycling method. However, green-synthesized NPs are preferred because they are eco-friendly, feasible, safe, cost-effective, use minimally toxic chemicals, and produce fewer harmful by-products. This method is preferred over physical and chemical methods because of its environmental benefits (Tajau et al. 2020). Over the years, multiple studies have investigated green synthesizing NPs. Dejene and Geletaw (2023b) examined green synthesizing ZnO NPs using additives from various natural source.

Methods of incorporating ZnO NPs in to polymer matrix

Nanobiocomposites (NBCs) are produced when nanoparticles are distributed in a biocomposite matrix for specific functionalization purposes. The nanoparticles used in NBCs, which are typically smaller than 100 nm, exhibit a better performance than traditional biocomposites. Studies have explored the use of different nanoparticles as fillers in NBCs for environmental sustainability. Polymeric NBCs can easily be reinforced with biofibers, and researchers have developed various processing techniques to achieve functional properties (Hasan, Horváth, and Alpár 2020). A potential mechanism for the formation of nano-biocomposites is illustrated in Figure 3.

ZnO NPs are commonly used in high-performance polymeric nanocomposites production, but uniform dispersion is crucial to prevent agglomeration due to surface forces. The three methods for producing defect-free high-quality polymeric nanocomposites (PNCs) filled with NPs include direct mixing, in situ polymerization, and solution-mixing techniques. Direct mixing methods, such as melt mixing, miscible solvent mixing, and powder metallurgy, are effective for preparing nanoparticles filled with PNCs (Hiremath et al. 2021). Most researchers have utilized direct mixing methods to mix ZnO NPs in natural-fiber-reinforced polymeric composites. Adlie et al. (2023) developed a foamed polymer composite material using an oil palm empty fruit bunch (OPEFB) fiber and ZnO, which was created through the pouring/casting method by pouring the mixture into a mold after thorough stirring in a mixing container. Dhanapal et al. (2023) used a miscible solvent mixing (acetone) process to study the effect of amine-functionalized nano-ZnO-reinforced areca fiber/epoxy hybrid nanocomposites, as shown in Figure 4. Ghalehno and Arabi (2021) studied the impact of nano-ZnO on the hygroscopic properties of PP/Wood flour composites. They used a melt mixing method in a twin-screw extruder with barrel temperatures ranging from 150°C to 170°C. After melting PP, maleic anhydride grafted polypropylene (PP-g-MA) and nano-ZnO were added, followed by the addition of wood floor, resulting in a total mixing time of 12 minutes.

Owing to their high surface-to-volume ratio, surface energy, and dielectric constant, ZnO NPs dispersions are prone to aggregation, which is generally detrimental to their performance and limits

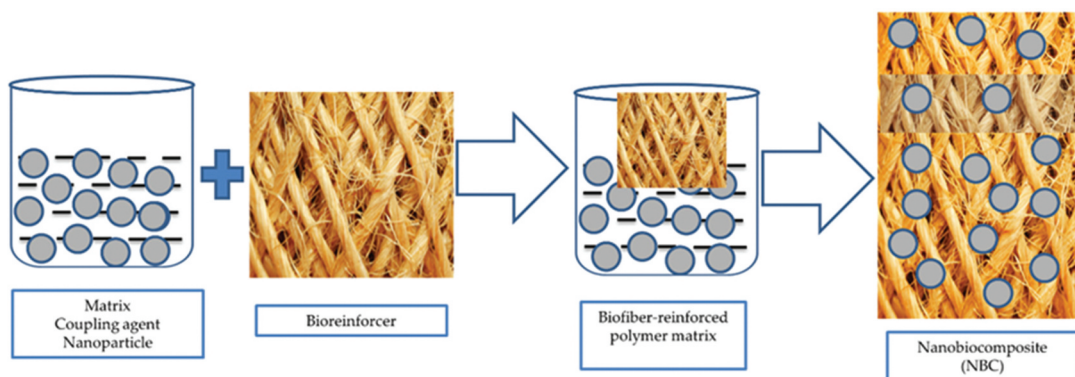


Figure 3. Formation mechanism of nano biocomposites (NBCs) (Hasan, Horváth, and Alpár 2020).

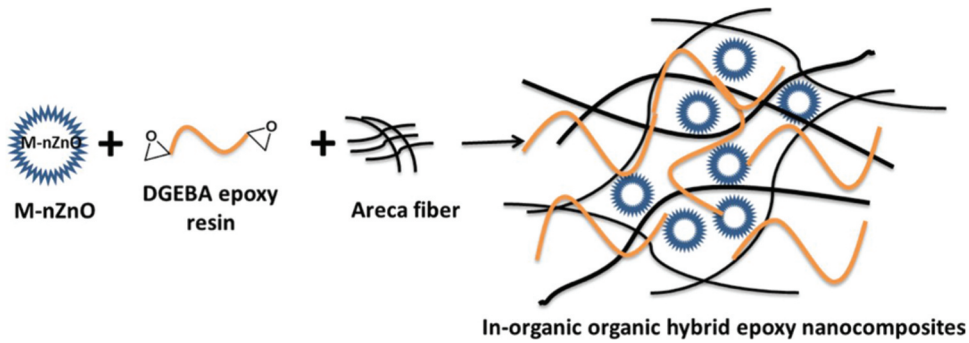


Figure 4. ZnO-nano material mixing and composite manufacturing process (Adlie et al. 2023; Dhanapal et al. 2023).

the application range of ZnO-based hybrids. Surface modification is essential to improve the dispersibility of ZnO NPs in a solution or composite matrix (Curri et al. 2003).

Methods to obtain uniform dispersion of NPs in polymer nanocomposites

Understanding the surface characteristics of reinforcing NPs is crucial for achieving a uniform dispersion and strong interfacial adhesion during the construction of polymeric matrices (Hiremath et al. 2021). Surface modification techniques are used to render inorganic nanofillers hydrophobic, decrease their hydrophilicity, and increase their hydrophobicity, thereby rendering them compatible with matrix materials. This approach helps overcome the challenges of achieving uniform dispersion in PNCs (Müller et al. 2017). Surface modification of inorganic NPs can be achieved either by physical or chemical interactions between NP and surface modifiers. Physical methods involve setting up forces, such as electrostatic, hydrogen bonding, and van der Waals forces, between the nanofillers and polymeric matrix material by coating the inorganic NPs with a surfactant with a low molecular weight or by a polymer with a reasonably high molecular weight. Physical techniques for nanoparticles have a major drawback in that polymeric coatings may be desorbed from particle surfaces. To overcome this, researchers have suggested the use of covalent bonds between the matrix and nanoparticles through chemical reactions or addition of coupling agents, which is widely used (Hiremath et al. 2021).

Salahuddin et al. (2017) found that even 0.8% ZnO nanotubes significantly improved the mechanical properties of epoxy PNCs, increasing the tensile strength by 27% and toughness by 105% compared with neat epoxy. Wong et al. (2014) investigated the impact of solvents on the nanoparticle dispersion, UV absorption resistance, and interfacial bonding in ZnO NPs. They fabricated glass-fiber-reinforced epoxy PNCs using isopropyl alcohol blending, revealing positive effects on NP dispersion and mechanical properties. However, one of the researches Wong et al. (2016) highlighted that silane surfactant treatment of ZnO NPs had no effect on UV absorption in the ZnO/DGEBA/micro-HGF composite. Research has shown that the addition of high amounts of ZnO NPs can induce steric hindrance, negatively affecting the curing behavior of epoxy PNCs. Additionally, more than 3.5 wt. % of ZnO NPs reduces reaction between resin and cross-linker, affecting PNC properties (Ramezanzadeh, Attar, and Farzam 2011). Devi and Maji (2012) created wood-fiber mat (*Bombex ceiba* L.) polymer nanocomposites by impregnating styrene-acrylonitrile copolymer, vinyl trichlorosilane (VTCS)-modified ZnO NPs, and glycidyl methacrylate (GMA) as a cross-linking agent.

Ultrasonication is another widely used technique for uniform NP dispersion in a matrix, which utilizes large energy transfers to disrupt the physical and chemical interactions between particles and composite phases (Hiremath et al. 2021). Murshid et al. (2016) examined the impact of ZnO and nanoclay on cross-linked jute-reinforced soy-flour green nanocomposites. The process involved ultrasonication to ensure good dispersion of ZnO and nanoclay particles within the resin matrix. Joshi et al. (2022) studied nano-ZnO-filled epoxy/basalt fiber composites, modifying epoxy resin with varying wt.% ZnO nanofiller, and achieving nanofiller dispersion through sonication technique. Hari et al. (2021)

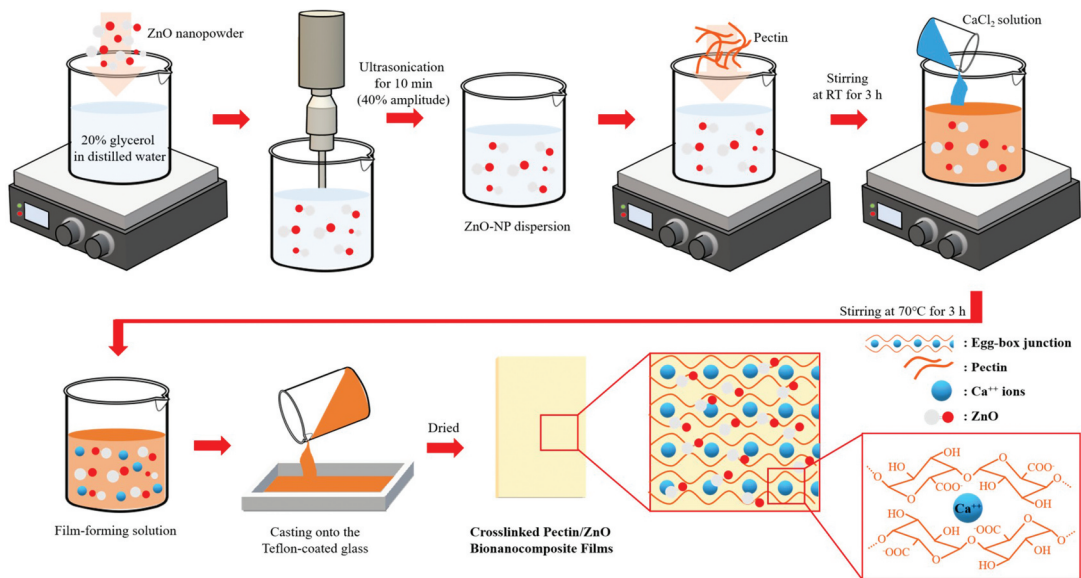


Figure 5. Ultrasonication and fabrication of crosslinked pectin/ZnO bionanocomposite films (Hari et al. 2021).

prepared crosslinked pectin/ZnO bionanocomposite films (CPZBF) by adding ZnO nanopowder to distilled water containing glycerol. The dispersion was ultrasonicated for 10 minutes, then 4g of pectin was dissolved in the dispersion and stirred for 3 hours. A CaCl_2 solution was added and mixed using a magnetic stirrer at 70°C for 20 minutes, as shown in Figure 5.

The impact of ZnO nanofillers on the properties of NFRCs

Nanocomposites are high-performance composite materials that are created by reinforcing nanofillers using advanced manufacturing methods. By integrating nanotechnology principles with fibers, fillers, and matrices, composite performance can be significantly improved. Studies have shown that nanofillers have high mechanical performance potential, making them suitable for various engineering applications. The quantity of nanofiller added to the polymer matrix depends on the filler type and the matrix used, with most studies ranging between 4% and 5% by weight (Ramesh et al. 2022). This section examines the impact of ZnO nanofiller materials on the properties of natural-fiber-reinforced polymer composites, focusing on their water repellency, mechanical properties, thermal stability, and antimicrobial activity, and discusses their applications.

Enhancing water repellency

Composite materials absorb moisture via diffusion, capillary action, and matrix microcracks (Kushwaha, Pandey, and Kumar 2014). Inorganic nanoparticles can increase the density of natural fiber composites by replacing low-density polymer resin with high-density nanoparticles, reducing the natural fiber content, and increasing the tensile strength (Mohammed, Rahman, Mohammed, Batar, et al. 2022). The use of nano-ZnO in natural fiber composites has been found to lead to lower water absorption compared to the control specimens. This is because nano-ZnO acts as an efficient barrier against moisture uptake, which increases the tortuosity of the water molecules diffusing through the composite panels. Additionally, nano-ZnO restricts the molecular mobility of polymer chains, making water penetration into the composite more difficult, potentially reducing the water effect (Ghalehno and Arabi 2021).

Figure 6 shows a schematic representation of the increased water repellency after incorporating NPs into NFRPCs.

Numerous researchers have investigated the impact of ZnO NPs on the water absorption properties of natural-fiber-reinforced composites. Ghalehno and Arabi (2021) created Polypropylene/wood flour composites with nano-ZnO content, resulting in reduced moisture absorption but improved dimensional stability, with the lowest water diffusion coefficient and swelling rate parameter. Adlie et al. (2023) found that nano-ZnO fillers can improve the hydrophobicity of oil palm fibers, increasing the contact angle by 34%. These fillers can diminish the water absorption properties and bind composite materials such as polyurethanes and oil palm empty fruit bunch (OPEFB) fibers. The addition of nano-ZnO decreases the void content of the composites, making water absorption more difficult. This is due to fewer hydrogen bonds between wood flour and water molecules, resulting in nanoparticle dispersion and agglomeration, which affects water absorption.

Surface modification of nanoparticles in polymer matrices is a common strategy for preventing nanoparticle agglomeration and improving composite hydrophobic properties. Devi and Maji (2012) developed wood-fiber mat-reinforced polymer nanocomposites (WPNC) by impregnating a styrene-acrylonitrile copolymer (SAN) in a 2:3 molar ratio, vinyl trichlorosilane (VTCS)-modified ZnO NPs, and glycidyl methacrylate (GMA) as a cross-linking agent. Vacuum impregnation improved the hydrophobic properties of the wood-fiber mat composites. Water uptake decreased with SAN copolymer impregnation, whereas the untreated samples absorbed more water. The SAN copolymer filled up void spaces in the wood, while the GMA-treated wood showed lower water absorption due to cross-links formed by the GMA interaction with the SAN copolymer and the hydroxyl group of the wood. The SEM images in Figure 7a show severe agglomeration owing to the high surface energy, whereas the VTCS-modified ZnO NPs showed better dispersibility, as depicted in Figure 7b. The SAN/ZnO-treated samples exhibited lower water absorption than that of the SAN-treated wood samples.

Enhancing mechanical properties

Polymer nanocomposites (PNCs) have become versatile materials for various technological applications including aerospace, automobiles, biomedical devices, electronics, and energy storage.

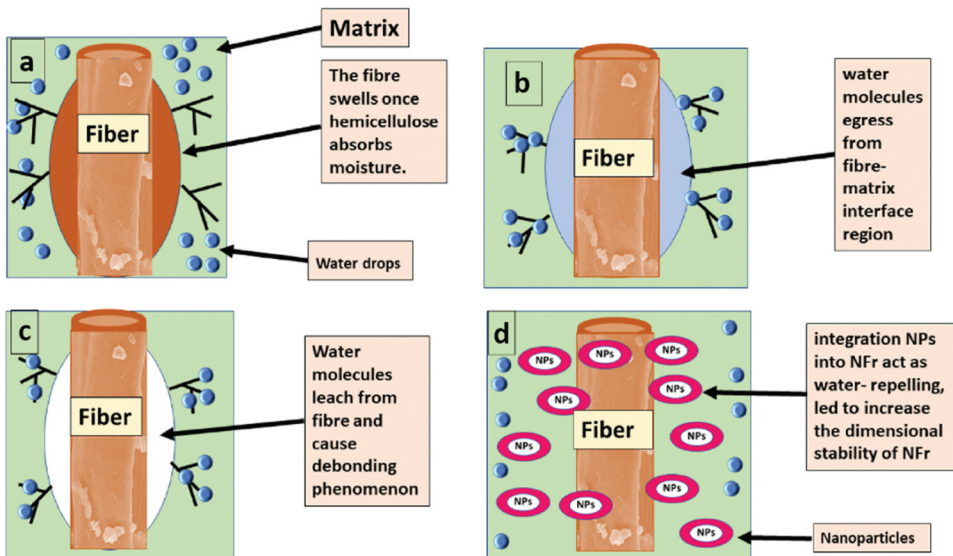


Figure 6. Schematic representation of increased water repellency after incorporating NPs into NFRPC (Mohammed, Rahman, Mohammed, Adam, et al. 2022).

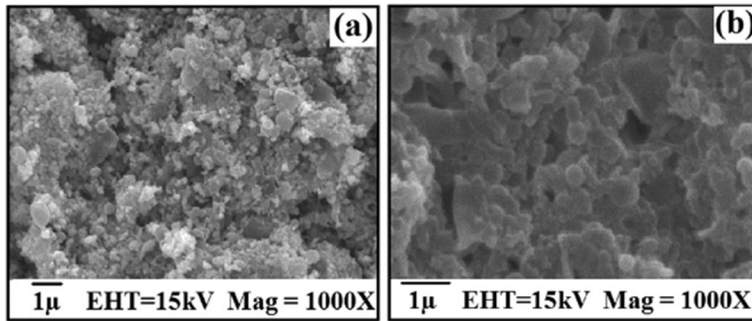


Figure 7. SEM image of (a) ZnO and (b) VTCS-modified ZnO (Devi and Maji 2012).

A thorough understanding of the effects of NPs on the mechanical properties of composites is imperative to ensure their reliability and performance. Researchers have proposed theories to elucidate the synergetic interaction between the polymeric matrix and the chemical and geometrical characteristics of NPs, as well as the interfacial interactions between the filler and the matrix. Interactions among nanoscale filler particles, polymer matrix materials, and additional substitute materials play a pivotal role in determining the ultimate mechanical properties of multiphase materials. Understanding the interaction mechanisms between different phases in PNCs is essential for identifying novel microstructures and multilevel complex stress transfers (Hiremath et al. 2021). According to Ramesh et al. (2022), incorporating filler elements in natural fibers and composites enhances mechanical properties, reduces water absorption, and improves interfacial adhesion, stress transfer, and stiffness. A higher filler content promotes better interactions between the particles and the matrix.

Numerous researchers have investigated the impact of ZnO nanofillers on natural-fiber-reinforced polymeric composites for advanced applications. Devaraju and Sivasamy (2018) investigated the development of sisal fiber (SF) composites activated with NPs using the sol-gel method. By incorporating 0.5 wt. % of NPs into distinct composites, they observed that the ZrO_2 -NPs SF composite exhibited superior mechanical properties compared to ZnO-NPs SF and conventional sisal fiber composites. The enhanced interlaminar strength and covalent bonding between the nanoparticles and sisal fiber matrix are key factors in the performance of the composites. In a related study, Devaraju et al. (2020) scrutinized the tensile, impact, and flexural properties of palm fiber composites and showed that the addition of 0.5 wt.% ZnO NPs improved their mechanical properties, highlighting the potential of NP-activated palm fiber composites.

Ghalehno and Arabi (2021) conducted a study on the development of polypropylene/wood flour composites with varying nano-ZnO content through the process of melt compounding and injection molding. The composites exhibited notable improvements in both tensile strength and modulus, reaching maximum values of 30.74 and 2571.69 MPa for composites filled with two parts per hundred of compound (2phc) (WF + PP) nano-ZnO, as depicted in Figure 8a. In addition, the impact of nano-ZnO on the flexural strength and modulus was most significant in specimens treated with 2phc nano-ZnO, showing values of 18.51 and 1623.77 MPa, respectively, compared with untreated specimens, as shown in Figure 8b. In a related study, Joshi et al. (2022) explored the fabrication and mechanical characterization of nano-ZnO-filled epoxy/basalt fiber composites. The epoxy resin was modified with ZnO nanofillers, dispersed by sonication, and fabricated through wet layup, followed by curing under compression molding. The most notable enhancement in strength was observed in the basalt-epoxy-zinc (BEZ2) composite containing 2 wt.% ZnO nanofillers, with flexural and tensile strength increasing by 22% and 9.78% compared to unfilled BEZ0 composite. However, over 2 wt.% ZnO nanofiller addition led to composite failure due to delamination, matrix cracking, and fiber fracture as depicted in Figure 8(c–f). Researchers have explored the use of woven fabric reinforcement compared to fibers in epoxy composites; hence, the addition of ZnO NPs with lower agglomeration and increased the

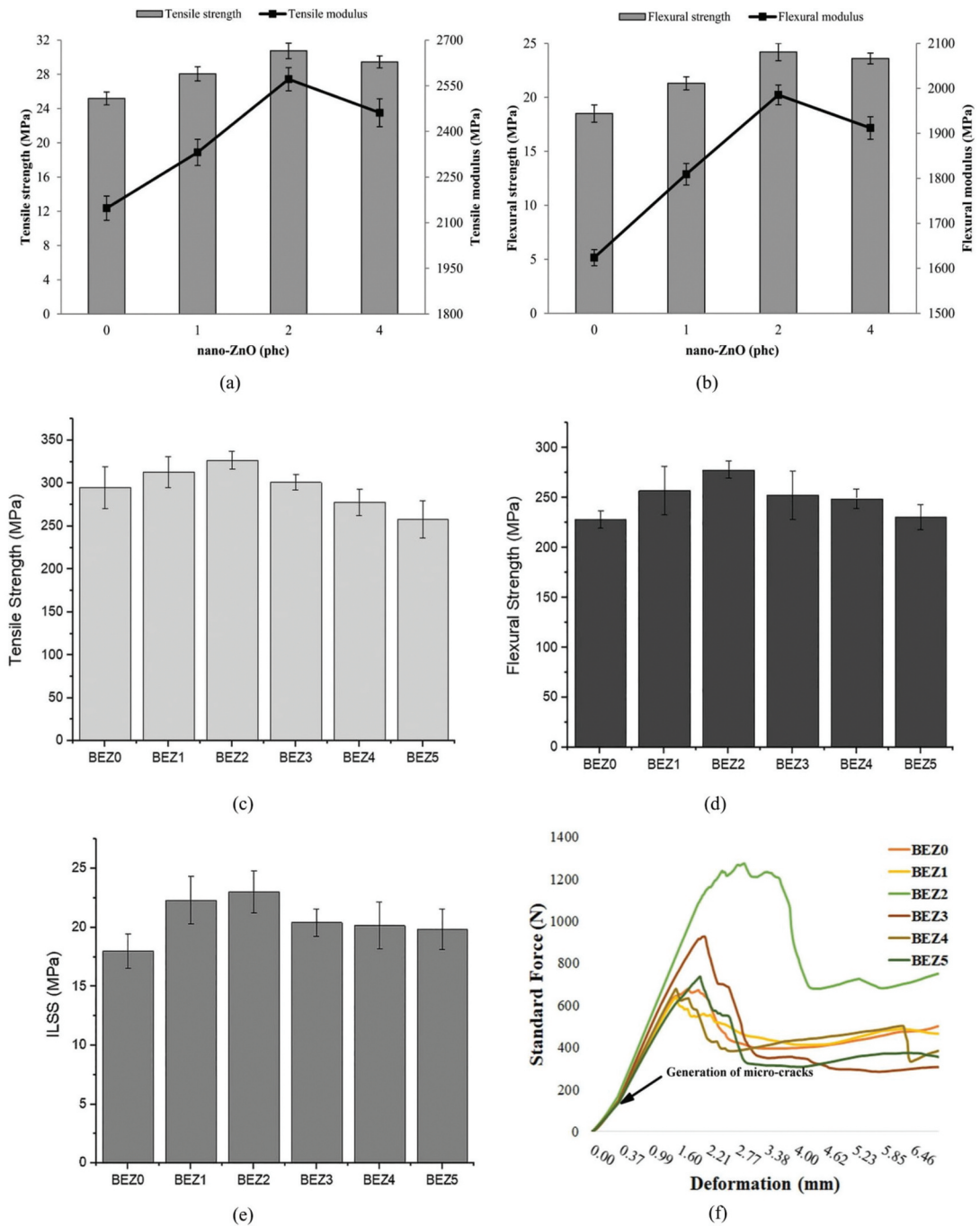


Figure 8. Tensile and flexural properties of the wood polymer composite reinforced with different levels of ZnO NPs (a,b) (Ghalehno and Arabi 2021). Tensile, flexural, inter-laminar shear strength, deformation (c-f) basalt fiber epoxy ZnO NPs (BEZ) (Joshi et al. 2022).

content of ZnO NPs in the woven fabric reinforcement, thereby enhancing the properties of the composites. Sathishkumar et al. (2022) improved the mechanical properties of jute fiber woven mat reinforced epoxy composites by adding zinc oxide filler. The results showed that the incorporation of

the zinc oxide filler gradually increased the mechanical strength and enhanced the bonding between the fiber and the matrix. Composites with 25% zinc oxide filler had the highest mechanical strength, whereas composites with more than 25% zinc oxide filler showed poor mechanical properties owing to the lower resin content and bonding between the fiber and matrix.

Nanosized inorganic particles in polymer matrices improve the mechanical properties of composites; however, agglomeration makes it difficult to achieve uniform dispersion. An increased ZnO loading restricts free movement, causing matrix yield and fiber pull-out (Joshi et al. 2022). The surface modification of ZnO NPs in polymers, often achieved using silane coupling agents or certain polymers, is a common strategy for preventing nanoparticle agglomeration. In a recent study by Dhanapal et al. (2023), the effect of amine-functionalized nano-ZnO-reinforcements on areca fiber/epoxy hybrid nanocomposites was investigated. The maximum mechanical performance of the Af/DGEBA epoxy resin was observed with incorporation of up to 2 wt. %, as shown in Figure 9. This enhancement can be attributed to the homogenous dispersion of the M-nZnO NPs in the DGEBA epoxy resin, which could only be achieved with up to 2 wt. % addition of M-nZnO. This facilitated stronger adhesion between the areca fiber and DGEBA epoxy, resulting in improved mechanical properties. Similarly, in a study conducted by Sagar et al. (2022), the addition of nano-ZnO to a jute fiber/epoxy composite enhanced wear resistance and compressive strength. The inclusion of ZnO led to a significant 43% increase in the compressive strength in the short-beam shear test and an impressive 80% reduction in the wear rate when 3 wt.% ZnO was added. This finding suggests that modified ZnO holds promise as a favorable nanofiller for reinforcing polymeric matrices integrated with natural fibers.

Hybrid nano-ZnO materials have been created and tested due to electrostatic interactions between nanoparticles and ZnO, enhancing dispersibility and providing new functions for the host polymer. Research has demonstrated that incorporating nanoclay, SiO₂, and ZnO into wood polymer composites can lead to improved mechanical performance and reduced water absorption capacity. Bajwa et al. (2021) investigated the thermal stability, dynamic mechanical properties, and flammability characteristics of a CNC/ZnO nanohybrid-filled PLA matrix, and found that the adsorption of nano-

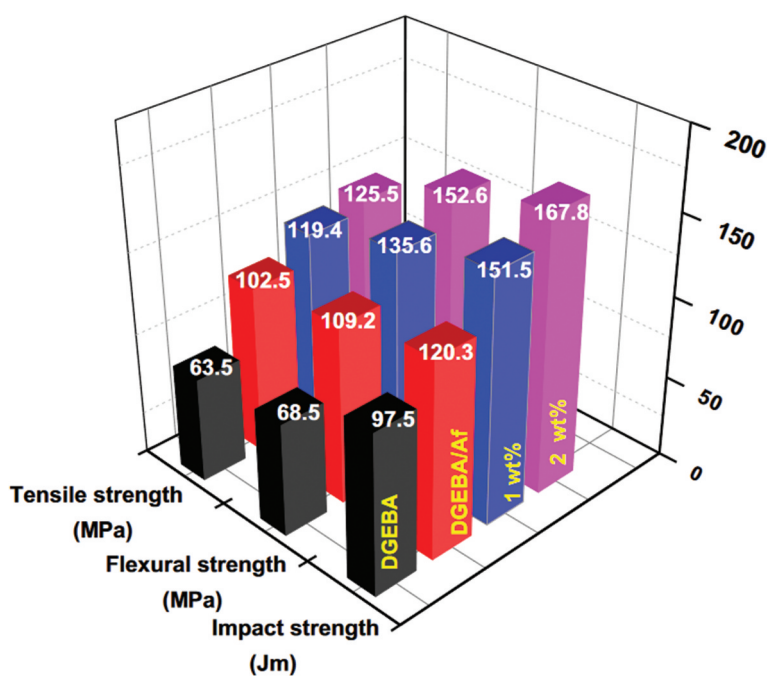


Figure 9. Mechanical properties of pure (DGEBA) epoxy matrix, amine-functionalized nano-ZnO-reinforced areca fiber/epoxy hybrid nanocomposites (Af/DGEBA) (Dhanapal et al. 2023). Reprinted with permission from springer nature.

ZnO particles improved the dispersion and increased the thermal properties of the host polymer. Another study by, Prasob and Sasikumar (2018) found that jute fiber reinforced nano composite specimens (ZnO and TiO₂) had enhanced mechanical strength characteristics compared to unfilled composites, with matrix toughening, microcrack bridging, and good interfacial bonding contributing to these improvements. Allamraju (2018) evaluated the strength of hybrid jute/glass fiber-reinforced epoxy composites filled with nanosized TiO₂ and ZnO fillers. The composites exhibited 60% and 71% higher compressive and impact properties, respectively, than the individual glass fiber composites, and the mechanical properties of the hybrid composites filled with ZnO outperformed those of the composites filled with TiO₂.

The sustainable use of biological waste is increasing, and various fiber sources are being studied for their application in various fields. In a recent study by Adlie et al. (2023), the effects of incorporating ZnO and polyurethane into oil palm empty fruit bunch-reinforced polymer composites for automotive interior components were investigated. The resulting composites exhibited increased tensile, compressive, flexural, and impact strengths as well as thermal stability. Similarly, Sridhar et al. (2019) investigated the impact of ZnO NPs on the mechanical properties of a randomly oriented Chicken Feather Fiber (CFF)-reinforced Vinyl Ester (VE) composite. Optimal mechanical properties were achieved through a specific combination of CFF length (6 mm) and content (20 wt. %), in addition to ZnO content (1.5 wt.%). Furthermore, Torres et al. (2022) explored the mechanical properties and fracture behavior of bio-based epoxy laminates reinforced with agave fibers and zinc oxide. This study included two primary agave species, Ixtle and Henequen, with biolaminates prepared using a vacuum-assisted resin infusion process. The results indicated that the Ixtle bio-laminates had poor mechanical properties at higher filler concentrations, whereas the Henequen bio-laminates showed better properties.

Enhancing thermal properties

Polymer matrix composites exhibit poor thermal conductivities, which hinders their use in heat exchangers and thermal storage devices. To address this limitation, researchers are actively engaged in enhancing thermal conductivity and stability through a detailed examination of the material properties and heat transfer kinetics at various scales ranging from micro to atomic levels (Hiremath et al. 2021). The complex chain structure of polymer materials causes gaps and discontinuities in the chain, which can be bridged by conductive nanofillers such as zinc-based fillers (Chen et al. 2016). These nanofillers play a pivotal role in enhancing the thermal conductivity and stability of nanocomposites, thereby attracting significant attention from both the industry and academia (Adlie et al. 2023).

Thermogravimetric analysis (TGA) is a crucial analytical tool for investigating the thermal degradation behavior of polymeric materials under high-temperature conditions and for assessing the weight loss resulting from volatile product formation. It helps understand the raw material and finished product behavior, with nano-additives potentially enhancing stability (Amjad et al. 2022). The research conducted by Adlie et al. (2023) focused on examining the impact of incorporating ZnO nanofillers and oil palm empty fruit bunches (OPEFB) on the thermal stability of polyurethane polymer composites. These findings indicated that the addition of ZnO to the polyurethane composite resulted in enhanced thermal stability. The best composition was specimens with 15% ZnO and 149 microns OPEFB fibers particle size, making it a promising candidate for automotive interior components. The heating process started at 30°C, resulting in a small mass loss owing to water evaporation. Degradation occurs at temperatures above 300°C, with the highest T_{onset} at 285.88°C for 0% ZnO. Maximum degradation occurs at 436.32°C with 15% ZnO, causing a weight loss of up to 76%. The composite without the ZnO filler had a maximum degradation temperature of 402.61°C and the highest mass reduction of 96.96%. The composite decomposes to produce a carbon residue at 460–600°C, indicating that the addition of ZnO enhances thermal stability, as shown in Figure 10a.

Surface modification or the incorporation of hybrid nanoparticles can mitigate nanoparticle agglomeration in polymer matrices. This can be effectively accomplished by using silane coupling agents or tailored polymers. Dhanapal et al. (2023) conducted a study that demonstrated the enhancement of thermal properties in areca fiber/epoxy hybrid nanocomposites through the incorporation of amine-surface-treated nano-ZnO. The research revealed that optimal performance was achieved with 2 wt.% M-nZnO NPs. In a separate investigation, Devi and Maji (2012) developed polymer nanocomposites using wood-fiber mat and a combination of styrene-acrylonitrile copolymer, vinyl trichlorosilane (VTCS)-modified ZnO NPs, and glycidyl methacrylate (GMA). The thermal stability of the composite samples treated with ZnO NPs was evaluated using TGA, as illustrated in Figure 10b. Recently, attention has been drawn toward nanocomposites, as they incorporate nano-sized particles into a continuous polymer host to improve their thermal properties, especially at low levels of nanosized fillers. A study conducted by Murshid et al. (2016) revealed that the incorporation of ZnO NPs into nanoclays reinforced with jute fabric and soy flour green nanocomposites improved the thermal stability of the synthesized nanocomposites. This enhancement was attributed to the heat-shielding effect of the ZnO NPs and restricted diffusion of volatile decomposition products within the nanocomposites. Moreover, Azizi et al. (2014) reported that cellulose nanocrystal/zinc oxide (CNC/ZnO) could be utilized as bifunctional fillers in a poly(vinyl alcohol) and chitosan matrix. The combination of CNCs and nano-ZnO significantly improved the tensile strength and modulus of PVA/Cs. The hybridization approach also highlighted the thermal stability owing to the strong interaction between the CNCs and ZnO, indicating the formation of a protective barrier against thermal decomposition. Bajwa et al. (2021) studied the impact of nano-ZnO/CNC particles on the thermal stability of the biodegradable polymer polylactic acid (PLA). Their research revealed that the adsorption of nano-ZnO/CNC particles improved dispersion in the PLA matrix and increased the thermal properties of the host polymer. Specifically, the incorporation of 1.5% nano-CNC-overlaid ZnO NPs into PLA enhanced its mechanical strength, thermal properties, and flame resistance. Additionally, the PLA formulation containing 1.5% CNCs demonstrated the highest storage modulus, whereas those with 1.5% CNCs and 2.5% ZnO displayed the lowest values, as depicted in Figure 10C.

Moreover, the incorporation of NPs within the polymer matrix has been shown to significantly influence the crystallization behavior and glass transition temperature of composite materials, resulting in improved thermal stability and flame-retardant properties, while also serving as an effective thermal insulator. In a study by Jose et al. (2014), the examination of cross-linked polyethylene/ZnO nanocomposites revealed that surface-modified ZnO played a vital role in accelerating crystallization process and exhibited heterogeneous nucleating capability. Similarly, research conducted by Lorero et al. (2020) on composites reinforced with recycled ZnO NPs from alkaline batteries demonstrated that the inclusion of ZnO particles led to an increase in glass transition temperatures (T_g) due to their

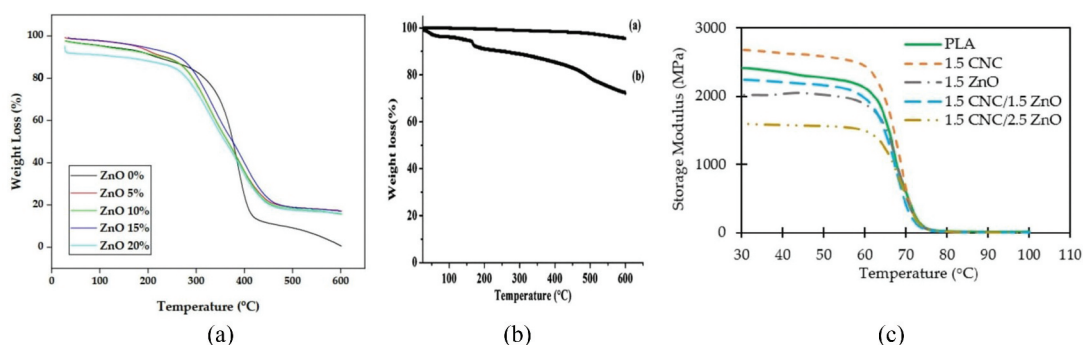


Figure 10. (A) TGA of oil palm empty fruit bunches composite at each variation of ZnO (Adlie et al. 2023). (B) TGA of (a) ZnO and (b) modified ZnO. Reprinted with permission from (Devi and Maji 2012). (C) Representative curve of storage modulus vs. temperature PLA and corresponding nanocomposites (Bajwa et al. 2021).

steric hindrance effects and their ability to catalyze curing through acid-base interactions with hardener amine groups. The presence of internal moisture within the recycled ZnO structure was found to enhance its interaction with the polymer matrix, subsequently enhancing its thermal stability, particularly at medium filler loads. Recently, Ponnammam et al. (2019) reported that ZnO NPs caused a reduction in the matrix glass transition temperature (T_g) and matrix crosslinking density owing to imperfect dispersion or aggregate formation. This study investigated the impact of filler content, size, matrix polymer, and interface chemistry on the nonlinear thermal properties of ZnO/polymer composites. Tian et al. (2022) study revealed that the increase in ZnO varistor particle size led to enhancement in dynamic mechanical properties and thermal stability in ZnO varistor-epoxy composites. This was attributed to the improved dispersion of large ZnO particles within the polymer matrix, resulting in higher stiffness and limited mobility of the motion unit.

Enhancing antimicrobial properties

ZnO NPs exhibit antimicrobial activity by producing reactive oxygen species (ROS), particularly hydrogen peroxide, in light. They have been proven effective against various microorganisms, including *B. subtilis*, *S. aureus*, *E. coli*, *Salmonella*, *P. aeruginosa*, and *C. jejuni*. Bacteria that were more sensitive to ZnO NPs showed a greater sensitivity to H_2O_2 . However, research on *S. aureus* has shown that the antimicrobial activity of ZnO NPs is independent of ROS production, suggesting an alternative mechanism that affects cell energy metabolism and amino acid biosynthesis. They can release cytotoxic Zn^{2+} ions upon partial dissolution (Omerović et al. 2021).

Researchers have explored the antimicrobial properties of ZnO NPs as a matrix filler. Dhanapal et al. (2023) improved the antimicrobial activity of composites by incorporating amine-functionalized nano-ZnO into the matrix. This study found that the addition of a 2 wt.% M-nZnO to the Af/DGEBA epoxy increased the zone of inhibition against pathogenic bacterial strains. This could be the first step in bacterial inactivation, leading to increased membrane permeability, permeation of cellular materials, and leakage of potassium ions. Proteins contain elements, such as nitrogen, phosphorus, and sulfur, which have a strong affinity for M-nZnO. This results in severe damage to the plasma membrane and cell wall of the bacterial cell by oxidizing lipids present in the membrane. Yu et al. (2015) prepared a CNC/ZnO nanohybrid composite, CNC/ZnO-0.5, which demonstrated high antibacterial activity against *S. aureus* and *E. coli* bacteria, as shown in Figure 11. The higher antibacterial ratio was due to the smaller ZnO NPs with narrow size distribution.

Armynah et al. (2022) explored the use of ZnO NPs and natural fiber as reinforcement for starch-based bioplastics. Researchers have found that ZnO NPs and natural fibers significantly enhance the mechanical, antibacterial, and physical properties of cassava starch/chitosan/pineapple leaf fiber (PALF)/ZnO bioplastic films. The results showed no fungal growth after 30 days of bioplastic coating with different percentages of ZnO NPs, indicating their significant role in bioplastic properties. However, ZnO NPs are commonly used in antimicrobial applications in nanocomposite films for food packaging, without the addition of natural fibers. Seray et al. (2021) developed poly (butylene adipate-co-terephthalate)/ZnO NPs films, ensuring controlled release of Zn^{2+} ions for safe food packaging. Novel nano-biocomposites combining ZnO NPs and essential oils have been developed in recent years. Wu et al. (2019) synthesized protein isolate-based biocomposite films using liquid precipitation to create antibacterial and antifungal materials. The composite films exhibited homogeneous, uniform, compact, and nonporous surfaces, which prevented light transmission, reduced oxidative deterioration, and extended the shelf life of the food. The antifungal activity of the film was higher than that of a pristine film. Antimicrobial composites have been created using ZnO NPs in biodegradable polymers such as polybutylene succinate (PBS) resin (Petchwattana et al. 2016). The antibacterial activity was highest in acidic environments, with the highest migration of ZnO NPs. Azizi et al. (2014) dispersed cellulose nanocrystals/zinc oxide (CNCs/ZnO) as bifunctional nano-sized fillers into PVA and chitosan matrix. The biocomposite films showed antibacterial activity against *Salmonella choleraesuis* and *Staphylococcus aureus*. Few studies have investigated the properties of

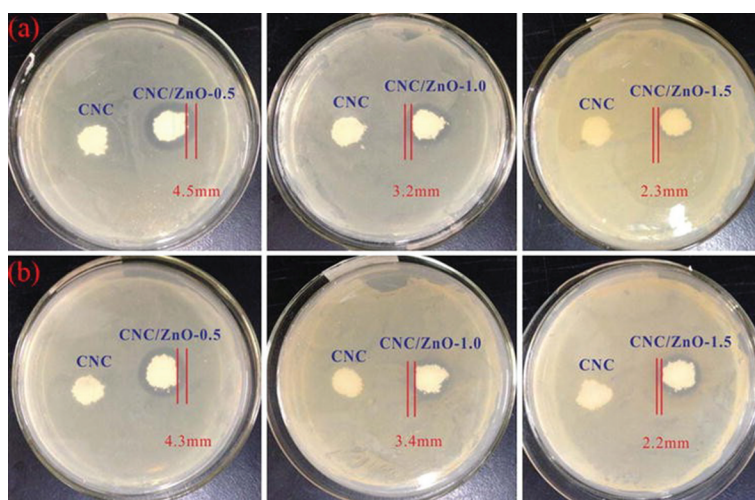


Figure 11. Antimicrobial activities of CNC/ZnO nanohybrids composites against *S. aureus* (a) and *E. coli* (b) (Yu et al. 2015).

ZnO NPs coatings on packaging to extend the shelf life of injera. Legassa et al. (2021) found that Ag NPs and ZnO NPs significantly increased shelf life up to 24 days at higher concentrations, but did not affect moisture or pH. Further research is needed to incorporate these nanoparticles and natural fibers into plastics to reduce migration issues and improve the durability of plastics.

Applications and future prospects

NFRCS with enhanced properties incorporating ZnO NPs as fillers can be utilized in various industries, offering innovative and sustainable solutions. (Table 3) summarizes the benefits of incorporating ZnO NPs as matrix fillers in natural-fiber-reinforced composites. Devaraju et al. (2020) studied the tensile, impact, and flexural properties of palm fiber composites and found them to be harmless to the environment, low-cost, and easily available, making them a potential wood substitute for indoor applications and automobile components. The study examines nano-ZnO content's impact on moisture absorption and dimensional stability in composites made from wood flour and polypropylene with a coupling agent, potentially serving as value-added materials (Ghalehno and Arabi 2021). Torres et al. (2022) studied the mechanical properties and fracture behavior of agave fiber bio-based epoxy laminates reinforced with zinc oxide. The results highlight the need to continue evaluating the potential applications of these green composites in the construction and automotive industries. Murshid et al. (2016) reported the effect of ZnO NPs alone and in combination with nanoclay reinforced with jute fabric and glutaraldehyde-crosslinked soy flour “green” nanocomposites. Consequently, the ZnO NPs and nanoclay-filled SF/J composites are eco-friendly and can be used for applications in new fields. Areca fiber-reinforced epoxy nanocomposites exhibit exceptional adhesion, excellent mechanical properties, resistance to water and antimicrobial activity, and are suitable for various engineering applications as protective polymer materials, enhancing performance and durability (Dhanapal et al. 2023). Numerous researchers have investigated the use of ZnO NPs as matrix fillers in food packaging applications to reduce the migration of fibers/packaging materials. The use of ZnO NPs in NFRCS presents promising opportunities for improved mechanical, thermal, antimicrobial, and sustainable properties, thereby enhancing durability and environmental sustainability. A compelling review conducted by Dejene (2024a) explored the application of ZnO NPs for the surface treatment of natural fibers prior to composite manufacturing, and revealed interesting results on the mechanical, thermal, and water resistance properties of their composites. Therefore, there is a clear need for further investigation to evaluate the effectiveness of utilizing ZnO NPs for fiber treatment compared to their incorporation as

Table 3. Summarized benefits of ZnO NPs incorporation in NFRCs as matrix filler.

Composite	Mixing method	Composite fabrication	Water absorption	Mechanical properties	Thermal stability	References
Areca fiber/epoxy ZnO NPs	Miscible solvent mixing	Hand layup	–	The introduction of M-nZnO improved the mechanical and thermal properties.	The thermal properties were most favorable with 2 wt.% M-nZnO NPs.	(Dhanapal et al. 2023)
Polypropylene/wood flour/nano-ZnO	Melt compounding	Injection molding	Nano-ZnO improved contact angle by 34% (95°) in untreated samples compared to 71°.	The maximum tensile strength and modulus values were 30.74 and 2571.69 MPa for composites filled with 2 phc nano-ZnO.	–	(Ghalehno and Arabi 2021)
Jute/soy flour/nano-ZnO /clay	Melt compounding	Compression molding	–	The composite with 5% ZnO NPs and 3% clay was found to exhibit enhanced mechanical properties.	The incorporation of ZnO NPs improved the thermal, flame retardancy, UV resistance, and dimensional stability.	(Murshid, Mandal, and Maji 2016)
OPFB/polyester/ZnO filler	Pouring/casting method	Hand lay-up	–	The results showed that adding ZnO and polyurethane to the composite increased tensile, compressive, flexural, and impact strength.	the results showed that adding ZnO and polyurethane to the composite increased thermal stability.	(Sathishkumar et al. 2022)
Wood/Styrene Acrylonitrile composite with ZnO NPs	Solution mixing	Hand lay-up	Wood samples treated with SAN/GMA/ZnO-VTCS showed improved dimensional stability and water resistance.	The flexural and mechanical properties were found to be improved due to inclusion of ZnO NPs into the wood.	The TGA analysis exhibited an improved thermal stability of wood samples treated with ZnO NPs.	(Devi and Maji 2012)
Wood/PE/PVC/clay/SiO ₂ /ZnO NPs	Melt mixing	Compression molding	Clay, SiO ₂ , and ZnO NPs significantly reduced water absorption capacity with 3 phr each.	Clay, SiO ₂ , and ZnO NPs significantly improved mechanical and hardness properties.	Clay, SiO ₂ , and ZnO enhance thermal properties with 3 phr each.	(Deka, Baishya, and Maji 2014)
Jute/Epoxy/ZnO/TiO ₂ NPs	Solvent mixing	Hand lay-up	The experimental results revealed that the 4 wt.% addition of TiO ₂ NPs reduces moisture content diffusion by 214%.	The maximum increase in tensile and compressive strength is 30.79% and 34.03% for TiO ₂ filled composite (4 wt.%) compared with the unfilled composite.	However, the glass transition temperature of the Jute/Epoxy nano composite has not been improved substantially.	(Prasob and Sasikumar 2018)
Poly (lactic acid) (PLA)/CNC/ZnO NPs	Solvent casting(chloroform)	Melt blending extrusion	–	The results demonstrated that the incorporation of 1.5% nano-CNC-overlaid ZnO NPs into PLA enhanced the mechanical characteristics.	The incorporation of 1.5% nano-CNC-overlaid ZnO NPs into PLA enhanced the thermal characteristics.	(Bajwa et al. 2021)
Chitosan/PVA/Cellulose nanocrystals/ZnO	Solvent mixing	Solvent casting	–	Tensile strength and modulus of bio-nanocomposite films increased from 55.0 to 153.2 MPa and from 395 to 932 MPa, respectively with increasing nano-sized filler amount from 0 to 5.0 wt. %.	The thermal stability of PVA/Cs was also enhanced at 1.0 wt. % CNCs/ZnO loading.	(Azizi et al. 2014)

fillers in the matrix for creating NFRCs. Furthermore, the incorporation of ZnO NPs as a filler, combined with cellulosic fibers sourced from agro-waste, in the naturally extracted thermoplastic starch matrix, is recommended for active packaging applications, as proposed by Dejene and Geletaw (2024).

Conclusion

The use of ZnO NPs as fillers in NFRCs can potentially improve their mechanical strength, water repellence, thermal stability, and antimicrobial properties. This could lead to new applications in automotive, construction, and packaging industries. Uniform dispersion of nanoparticles is crucial for quality composite fabrication, and surface modification can reduce the nanoparticle surface energy and prevent agglomeration. This uniform distribution provides stiffness to the matrix material and enhances the mechanical and thermal properties of composites. However, further research is needed to optimize the dispersion and alignment of ZnO NPs, investigate their long-term durability, and assess their environmental impacts. Overall, this approach contributes to the development of high-performance, sustainable materials with improved properties. In the future, it may be possible to work on hybridization by incorporating several types of NPs and analyzing the combined benefits of such mixing. Long-term durability and aging studies would also be beneficial for comprehending the potential of these ZnO NPs-reinforced natural fiber composites to withstand various environmental stressors, such as UV exposure, humidity, and thermal cycling. From a sustainability standpoint, additional research is necessary to investigate the biodegradability and recyclability of these composite materials as well as their potential for circular economy applications. The development of scalable manufacturing processes to produce larger quantities is also an important step towards commercialization. Finally, application-specific studies and computational modeling can provide in-depth insights into the structure-property relationships and failure mechanisms, ultimately guiding the design of ZnO NPs-reinforced natural fiber composites for specific end-use requirements.

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References

- Abdullah, A., S. A. Khalaf, J. Zaraket, and C. Salame. 2017. "Synthesis of ZnO Nanopowders by Using Sol-Gel and Studying Their Structural and Electrical Properties at Different Temperature." *Energy Procedia* 119. Elsevier:565–570. <https://doi.org/10.1016/j.egypro.2017.07.080>.

- Adib, B., S. M. M. I. Mahima Haque, K. M. Rafi Uddin, and K. M. R. Uddin Labib. 2024. "Heliyon Nanotechnology-Enhanced Fiber-Reinforced Polymer Composites: Recent Advancements on Processing Techniques and Applications." *Heliyon* 10 (2). Elsevier Ltd:e24692. <https://doi.org/10.1016/j.heliyon.2024>.
- Adlie, T. A., N. Ali, S. Huzni, I. Ikramullah, and S. Rizal. 2023. "Impact of Zinc Oxide Addition on Oil Palm Empty Fruit Bunches Foamed Polymer Composites for Automotive Interior Parts." *Polymers* 15 (2): 422. <https://doi.org/10.3390/polym15020422>.
- Akter, M., M. Haris Uddin, and H. Rahman Anik. 2024. "Plant Fiber-Reinforced Polymer Composites: A Review on Modification, Fabrication, Properties, and Applications." *Polymer Bulletin* 81 (1). Springer:1–85. <https://doi.org/10.1007/s00289-023-04733-5>.
- Allamraju, K. V. 2018. "Study of Mechanical Behaviour of Hybrid Jute Nano Fiber Composite." *Materials Today: Proceedings* 5 (9). Elsevier:20750–20753. <https://doi.org/10.1016/j.matpr.2018.06.460>.
- Amjad, A., A. Anjang Ab Rahman, H. Awais, M. Shukur Zainol Abidin, and J. Khan. 2022. "A Review Investigating the Influence of Nanofiller Addition on the Mechanical, Thermal and Water Absorption Properties of Cellulosic Fibre Reinforced Polymer Composite." *Journal of Industrial Textiles* 51 (1): 65S–100S. <https://doi.org/10.1177/15280837211057580>.
- Armynah, B., R. Anugrahwidya, and D. Tahir. 2022. "Composite Cassava Starch/Chitosan/Pineapple Leaf Fiber (PALF)/Zinc Oxide (ZnO): Bioplastics with High Mechanical Properties and Faster Degradation in Soil and Seawater." *International Journal of Biological Macromolecules* 213:814–823. <https://doi.org/10.1016/j.ijbiomac.2022.06.038>.
- Ashok, K. G., K. Kalaichelvan, and A. Damodaran. 2022. "Effect of Nano Fillers on Mechanical Properties of Luffa Fiber Epoxy Composites Effect of Nano Fillers on Mechanical Properties of Luffa Fiber Epoxy." *Journal of Natural Fibers* 19 (4). Taylor & Francis:1472–1489. <https://doi.org/10.1080/15440478.2020.1779898>.
- Azizi, S., M. B. Ahmad, N. Azowa Ibrahim, M. Zobir Hussein, and F. Namvar. 2014. "Cellulose Nanocrystals/ZnO As a Bifunctional Reinforcing Nanocomposite for Poly (Vinyl Alcohol)/Chitosan Blend Films: Fabrication, Characterization and Properties." *International Journal of Molecular Sciences* 15 (6). MDPI:11040–11053. <https://doi.org/10.3390/ijms150611040>.
- Baghdadi, Y. N., L. Youssef, K. Bouhadir, D. Patra, S. Mustapha, D. Patra, and A. R. Tehrani-Bagha. 2020. "The Effects of Modified Zinc Oxide Nanoparticles on the Mechanical/Thermal Properties of Epoxy Resin, *Journal of Applied Polymer Science* no. December 2019: 137 (43): 1–14. <https://doi.org/10.1002/app.49330>.
- Bag, D. S., A. Kumar, J. Kr Bansiwai, D. N. T. Gobardhan Lal, and E. Prasad Namburi. 2024. "Polymer Matrix Composites (PMCs) for Defence Applications." In *Novel Defence Functional and Engineering Materials (NDFEM)*, edited by E. P. Namburi, R. J. H. Wanhill, and D. K. Setua, 1–37. Vol. 2. Singapore: Springer. https://doi.org/10.1007/978-981-99-9795-4_1.
- Bajwa, D. S., J. Shojaeiarani, J. D. Liaw, and S. G. Bajwa. 2021. "Role of Hybrid Nano-Zinc Oxide and Cellulose Nanocrystals on the Mechanical, Thermal, and Flammability Properties of Poly (Lactic Acid) Polymer." *Journal of Composites Science* 5 (2): 43. <https://doi.org/10.3390/jcs5020043>.
- Balla, V. K., K. H. Kate, J. Satyavolu, P. Singh, and J. Ganesh Dattatreya Tadimet. 2019. "Additive Manufacturing of Natural Fiber Reinforced Polymer Composites: Processing and Prospects." *Composites Part B: Engineering* 174. Elsevier Ltd:106956. <https://doi.org/10.1016/j.compositesb.2019.106956>.
- Chen, H., V. V. Ginzburg, J. Yang, Y. Yang, W. Liu, Y. Huang, L. Du, and B. Chen. 2016. "Thermal Conductivity of Polymer-Based Composites: Fundamentals and Applications." *Progress in Polymer Science* 59. Elsevier:41–85. <https://doi.org/10.1016/j.progpolymsci.2016.03.001>.
- Curri, M. L., R. Comparelli, P. D. Cozzoli, G. Mascolo, and A. Agostiano. 2003. "Colloidal Oxide Nanoparticles for the Photocatalytic Degradation of Organic Dye." *Materials Science & Engineering: C* 23 (1–2). Elsevier:285–289. [https://doi.org/10.1016/S0928-4931\(02\)00250-3](https://doi.org/10.1016/S0928-4931(02)00250-3).
- Dahmardeh Ghalehno, M., and M. Arabi. 2021. "A Study on the Effect of Nano-ZnO on Hygroscopic Characteristics of PP/Wood Flour Composites." *Plastics, Rubber & Composites* 50 (10). Taylor & Francis: 516–523. <https://doi.org/10.1080/14658011.2021.1931768>.
- Dejene, B. K. 2024a. "Exploring the Potential of ZnO Nanoparticle- Treated Fibers in Advancing Natural Fiber Reinforced Composites: A Review." *Journal of Natural Fibers* 21 (1). Taylor & Francis. <https://doi.org/10.1080/15440478.2024.2311304>.
- Dejene, B. K. 2024b. "False Banana (Enset Ventricosum) Fibers: An Emerging Natural Fiber with Distinct Properties, Promising Potentials, Challenges and Future Prospects–A Critical Review." *Journal of Natural Fibers* 21 (1). Taylor & Francis. <https://doi.org/10.1080/15440478.2024.2311303>.
- Dejene, B. K., and T. M. Geletaw. 2024. "Development of Fully Green Composites Utilizing Thermoplastic Starch and Cellulosic Fibers from Agro-Waste: A Critical Review." *Polymer-Plastics Technology & Materials* 63 (5): 540–569. <https://doi.org/10.1080/25740881.2023.2296649>.
- Dejene, B. K., and T. Mamo Geletaw. 2023a. "A Review on False Banana (Enset Ventricosum) Fiber Reinforced Green Composite and Its Applications." *Journal of Natural Fibers* 20 (2). Taylor & Francis. <https://doi.org/10.1080/15440478.2023.2244163>.
- Dejene, B. K., and T. Mamo Geletaw. 2023b. "A Review of Plant-Mediated Synthesis of Zinc Oxide Nanoparticles for Self-Cleaning Textiles." *Research Journal of Textile and Apparel*. <https://doi.org/10.1108/RJTA-12-2022-0154>.

- Deka, B. K., P. Baishya, and T. K. Maji. 2014. "Synergistic Effect of SiO₂, ZnO and Nanoclay on Mechanical and Thermal Properties of Wood Polymer Nanocomposite." *Journal of Thermoplastic Composite Materials* 27 (4): 464–480. <https://doi.org/10.1177/0892705712452739>.
- Devaraju, A., and P. Sivasamy. 2018. "Comparative Analysis of Mechanical Characteristics of Sisal Fibre Composite with and without Nano Particles." *Materials Today: Proceedings* 5 (6). Elsevier Ltd:14362–14366. <https://doi.org/10.1016/j.matpr.2018.03.020>.
- Devaraju, A., P. Sivasamy, and G. Babu Loganathan. 2020. "Mechanical Properties of Polymer Composites with ZnO Nano-Particle." *Materials Today: Proceedings* 22. Elsevier Ltd:531–534. <https://doi.org/10.1016/j.matpr.2019.08.146>.
- Devi, R. R., and T. K. Maji. 2012. "Effect of Nano-ZnO on Thermal, Mechanical, UV Stability, and Other Physical Properties of Wood Polymer Composites." *Industrial & Engineering Chemistry Research* 51 (10). ACS Publications:3870–3880. <https://doi.org/10.1021/ie2018383>.
- Dhanapal, D., S. M. Suresh Kumar, J. M. Manjumeena Rajarathinam, A. Kumar Srinivasan, and A. K. Srinivasan. 2023. "Improvement of Physical–Mechanical–Antimicrobial Activities with Effect of Amine-Functionalized Nano-ZnO-Reinforced Areca Fibre/Epoxy Hybrid Nanocomposites." *Polymer Bulletin* 81 (5). Springer Berlin Heidelberg. 4369–4385. <https://doi.org/10.1007/s00289-023-04912-4>.
- Hari, K. D., C. V. Garcia, G. Shin, and J. Kim. 2021. "Improvement of the UV Barrier and Antibacterial Properties of Crosslinked Pectin/Zinc Oxide Bionanocomposite Films." *Polymers* 13 (15): 2403. <https://doi.org/10.3390/polym13152403>.
- Hasan, K. M. F., P. György Horváth, and T. Alpár. 2020. "Potential Natural Fiber Polymeric Nanobiocomposites: A Review." *Polymers* 12 (5): 1072. <https://MDPI/1072>.
- Hiremath, A., A. Amar, S. T. Murthy, K. N. Bharath, and I. P. Jones. 2021. "Nanoparticles Filled Polymer Nanocomposites: A Technological Review." *Cogent Engineering* 8 (1). Cogent. <https://doi.org/10.1080/23311916.2021.1991229>.
- Jose, J. P., L. Chazeau, K. T. V. Jean-Yves Cavaillé, S. Thomas, and S. Thomas. 2014. "Nucleation and Nonisothermal Crystallization Kinetics in Cross-Linked Polyethylene/Zinc Oxide Nanocomposites." *RSC Advances* 4 (60). Royal Society of Chemistry:31643–31651. <https://doi.org/10.1039/C4RA03731F>.
- Joshi, S., A. Hiremath, S. Yeshwant Nayak, J. Pramod Jaideep, and S. Thipperudrappa. 2022. "Hybridization Effect on the Mechanical Properties of Basalt Fiber Reinforced ZnO Modified Epoxy Composites." *Polymer Composites* 43 (8): 5704–5714. <https://doi.org/10.1002/pc.26889>.
- Kausar, A. 2021. "Progress in Green Nanocomposites for High-Performance Applications." *Materials Research Innovations* 25 (1). Taylor & Francis:53–65. <https://doi.org/10.1080/14328917.2020.1728489>.
- Kushwaha, P. K., C. N. Pandey, and R. Kumar. 2014. "Study on the Effect of Carbon Nanotubes on Plastic Composite Reinforced with Natural Fiber." *Journal of the Indian Academy of Wood Science* 11. Springer: (1): 82–86. <https://doi.org/10.1007/s13196-014-0121-3>.
- Legassa, O., A. Zewdu, and E. Mulugeta. 2021. "Improvement of Injera Shelf Life Through the Use of Silver (Ag) and Zinc Oxide (ZnO)." *Nano Particles* 11 (2): 52–64. <https://doi.org/10.5923/j.food.20211102.03>.
- Li, M., Y. Pu, V. M. Thomas, C. G. Yoo, S. Ozcan, Y. Deng, K. Nelson, and A. J. Ragauskas. 2020. "Recent Advancements of Plant-Based Natural Fiber-Reinforced Composites and Their Applications." *Composites Part B Engineering* 200:108254. <https://doi.org/10.1016/j.compositesb.2020.108254>.
- Lorero, I., M. Campo, G. Del Rosario, F. Antonio López, and S. González Prolongo. 2020. "New Manufacturing Process of Composites Reinforced with ZnO Nanoparticles Recycled from Alkaline Batteries." *Polymers* 12 (7): 17–21. <https://doi.org/10.3390/polym12071619>.
- Matysiak, W., T. Tański, and M. Zaborowska. 2019. "Manufacturing Process and Characterization of Electrospun PVP/ZnO NPs Nanofibers." *Bulletin of the Polish Academy of Sciences, Technical Sciences* 67 (2): 193–200. <https://doi.org/10.24425/bpas.2019.128601>.
- Mishra, T., P. Mandal, A. Kumar Rout, and D. Sahoo. 2022. "A State-Of-The-Art Review on Potential Applications of Natural Fiber-Reinforced Polymer Composite Filled with Inorganic Nanoparticle." *Composites Part C Open Access* 9 (July). Elsevier B.V.:100298. 100298. <https://doi.org/10.1016/j.jcomc.2022.100298>.
- Mochane, M. J., T. C. Mokhena, T. H. Mokhothu, A. Mtibe, E. R. Sadiku, S. S. Ray, I. D. Ibrahim, and O. O. Daramola. 2019. "Recent Progress on Natural Fiber Hybrid Composites for Advanced Applications: A Review." *Express Polymer Letters* 13 (2): 159–198. <https://doi.org/10.3144/expresspolymlett.2019.15>.
- Mohammed, M., R. Rahman, A. M. Mohammed, T. Adam, B. O. Betar, A. F. Osman, and O. S. Dahham. 2022. "Surface Treatment to Improve Water Repellence and Compatibility of Natural Fiber with Polymer Matrix: Recent Advancement." *Polymer Testing* 115 (July). Elsevier Ltd:107707. 107707. <https://doi.org/10.1016/j.polymertesting.2022.107707>.
- Mohammed, M., R. Rahman, A. M. Mohammed, B. O. Betar, A. F. Osman, T. Adam, O. S. Dahham, and S. C. B. Gopinath. 2022. "Improving Hydrophobicity and Compatibility Between Kenaf Fiber and Polymer Composite by Surface Treatment with Inorganic Nanoparticles." *Arabian Journal of Chemistry* 15 (11). The Authors:104233.104233. <https://doi.org/10.1016/j.arabjc.2022.104233>.
- Müller, K., E. Bugnicourt, M. Latorre, M. Jorda, Y. Echegoyen Sanz, J. M. Lagaron, O. Miesbauer, et al. 2017. "Review on the Processing and Properties of Polymer Nanocomposites and Nanocoatings and Their Applications in the

- Packaging, Automotive and Solar Energy Fields.” *Nanomaterials: Overview and Historical Perspectives* 7 (4). MDPI:74.74. <https://doi.org/10.3390/nano7040074>.
- Murshid, I., M. Mandal, and T. K. Maji. 2016. “Effect of Zinc Oxide and Nanoclay on Crosslinked Jute-Reinforced Soy Flour Green Nanocomposites.” *Journal of Composite Materials* 50 (6): 723–737. <https://doi.org/10.1177/0021998315580831>.
- Nayak, S., R. Kumar Nayak, and I. Panigrahi. 2021. “Effect of Nano-Fillers on Low-Velocity Impact Properties of Synthetic and Natural Fibre Reinforced Polymer Composites- a Review.” *Advances in Materials and Processing Technologies*. Taylor & Francis:1–24. <https://doi.org/10.1080/2374068X.2021.1945293>.
- Neto, J. S. S., H. F. M. de Queiroz, R. A. A. Aguiar, and M. D. Banea. 2021. “A Review on the Thermal Characterisation of Natural and Hybrid Fiber Composites.” *Polymers* 13 (24): 4425. <https://doi.org/10.3390/polym13244425>.
- Omerović, N., M. Djisalo, K. Živojević, M. Mladenović, J. Vunduk, I. Milenković, N. Knežević, I. Gadjanski, and J. Vidić. 2021. “Antimicrobial Nanoparticles and Biodegradable Polymer Composites for Active Food Packaging Applications.” *Comprehensive Reviews in Food Science and Food Safety* 20 (3): 2428–2454. <https://doi.org/10.1111/1541-4337.12727>.
- Petchwattana, N., S. Covavisaruch, S. Wiboonanawong, and P. Naknaen. 2016. “Antimicrobial Food Packaging Prepared from Poly (Butylene Succinate) and Zinc Oxide.” *Measurement* 93. Elsevier:442–448. <https://doi.org/10.1016/j.measurement.2016.07.048>.
- Ponnamma, D., J.-J. Cabibihan, S. S. P. Mariappan Rajan, K. Deshmukh, S. K. K. P. Jyoti Prasad Gogoi, M. Basheer Ahamed, B. N. C. Jagadish Krishnegowda, et al. 2019. “Synthesis, Optimization and Applications of ZnO/Polymer Nanocomposites.” *Materials Science & Engineering: C* 98. Elsevier:1210–1240. <https://doi.org/10.1016/j.msec.2019.01.081>.
- Prasob, P. A., and M. Sasikumar. 2018. “Static and Dynamic Behavior of Jute/Epoxy Composites with ZnO and TiO₂ Fillers at Different Temperature Conditions.” *Polymer Testing* 69. Elsevier Ltd.: 52–62. <https://doi.org/10.1016/j.polymertesting.2018.04.040>.
- Quadri, T. W., L. O. Olasunkanmi, O. E. Fayemi, M. M. Solomon, and E. E. Ebenso. 2017. “Zinc Oxide Nanocomposites of Selected Polymers: Synthesis, Characterization, and Corrosion Inhibition Studies on Mild Steel in HCl Solution” *ACS Omega* 2 (11). ACS Publications:8421–8437. <https://doi.org/10.1021/acsomega.7b01385>.
- Radhakrishnan, S., A. Khan, S. Prakash, D. Bhasha, S. Sumit, G. Pallav, and V. Chaudhary. 2023. “Studies on Mechanical, Thermal, and Water Immersion of Plant and Animal Wastage Nanofiller-Based Bio-Fiber-Reinforced Composites.” *Biomass Conversion and Biorefinery* (123456789). Springer Berlin Heidelberg. <https://doi.org/10.1007/s13399-023-04788-4>.
- Ramesh, M., L. Narasimhan Rajeshkumar, N. Srinivasan, D. Vasanth Kumar, and D. Balaji. 2022. “Influence of Fiber Material on Properties of Fiber-Reinforced Polymer Composites: A Review.” *E-Polymers* 22 (1): 898–916. <https://doi.org/10.1515/epoly-2022-0080>.
- Ramezanzadeh, B., M. M. Attar, and M. Farzam. 2011. “Effect of ZnO Nanoparticles on the Thermal and Mechanical Properties of Epoxy-Based Nanocomposite.” *Journal of Thermal Analysis and Calorimetry* 103 (2). Akadémiai Kiadó, co-published with Springer Science+ Business Media BV: 731–739. <https://doi.org/10.1007/s10973-010-0996-1>.
- Rashid, A. B., S. M. M. I. Mahima Haque, K. M. Rafi Uddin Labib, and K. M. R. Uddin Labib. 2024. “Nanotechnology-Enhanced Fiber-Reinforced Polymer Composites: Recent Advancements on Processing Techniques and Applications.” *Heliyon*. Elsevier. 10 (2): e24692. <https://doi.org/10.1016/j.heliyon.2024.e24692>.
- Sagar, K. G., K. Ramachandran, K. Kosanam, B. M. R. Bharathi, M. Rama, and S. Bhattacharya. 2022. “Analysis of Wear Behavior and Shear Properties of Nano-ZnO/jute Fiber/Epoxy Composites by Hand Layup Technique.” *Materials Today: Proceedings* 69. Elsevier:1274–1279. <https://doi.org/10.1016/j.matpr.2022.08.326>.
- Salahuddin, N. A., M. El-Kemary, and E. M. Ibrahim. 2017. “High-Performance Flexible Epoxy/ZnO Nanocomposites with Enhanced Mechanical and Thermal Properties.” *Polymer Engineering & Science* 57 (9). Wiley Online Library:932–946. <https://doi.org/10.1002/pen.24520>.
- Sathishkumar, T. P., L. Rajeshkumar, G. Rajeshkumar, M. R. Sanjay, S. Siengchin, N. Thakrishnan, M. Sriariyanun, et al. 2022. “Improving the Mechanical Properties of Jute Fiber Woven Mat Reinforced Epoxy Composites with Addition of Zinc Oxide Filler.” *In E3S Web of Conferences* 355:2006. <https://doi.org/10.1051/e3sconf/202235502006>.
- Seray, M., A. Skender, and A. Siham Hadj-Hamou. 2021. “Kinetics and Mechanisms of Zn²⁺ Release from Antimicrobial Food Packaging Based on Poly (Butylene Adipate-Co-Terephthalate) and Zinc Oxide Nanoparticles.” *Polymer Bulletin* 78. Springer: (2): 1021–1040. <https://doi.org/10.1007/s00289-020-03145-z>.
- Serkan, M. 2019. “Advancements in Properties of Cements Containing Pulverised Fly Ash and Nanomaterials by Blending and Ultrasonication Method (Review- Part II).” *Nano Hybrids and Composites* 24 (Table 1): 37–44. <https://doi.org/10.4028/www.scientific.net/NHC.24.37>.
- Shankar, S., L. Feng Wang, and J. Whan Rhim. 2018. “Incorporation of Zinc Oxide Nanoparticles Improved the Mechanical, Water Vapor Barrier, UV-Light Barrier, and Antibacterial Properties of PLA-Based Nanocomposite Films.” *Materials Science & Engineering: C* 93 (August). Elsevier:289–298. <https://doi.org/10.1016/j.msec.2018.08.002>.
- Siddiqui, M. A. S., M. S. Rabbi, and M. Tauhidur Rahman. 2024. “An Experimental Review on Enhancing the Low and High-Velocity Impact Resistance of NFCs Through Nanoparticle Reinforcement.” *Journal of Reinforced Plastics & Composites*. SAGE Publications Sage UK: London, England, 07316844241240569. <https://doi.org/10.1177/07316844241240569>.

- Sridhar, R., A. Athijayamanib, A. Alavudeenc, and K. Vinayagar. 2019. "Parametric Optimization of Effect of ZnO Nano Particle on the Mechanical Properties of Randomly Oriented Chicken Feather Fiber-Reinforced Vinyl ESTER COMPOSITE." *Journal of the Balkan Tribological Association* 25 (3): 767–778.
- Sridhar, J., R. Gobinath, and M. Serkan Kirgiz. 2022. "Comparative Study for Efficacy of Chemically Treated Jute Fiber and Bamboo Fiber on the Properties of Reinforced Concrete Beams." *Journal of Natural Fibers* 19 (15). Taylor & Francis:12224–12234. <https://doi.org/10.1080/15440478.2022.2054894>.
- Tajau, R., R. Rohani, S. Selina Abdul Hamid, Z. Adam, S. Najila Mohd Janib, and M. Zah Salleh. 2020. "Surface Functionalisation of Poly-APO-B-Polyol Ester Cross-Linked Copolymers As Core-Shell Nanoparticles for Targeted Breast Cancer Therapy." *Scientific Reports* 10 (1). Nature Publishing Group UK London:21704. <https://doi.org/10.1038/s41598-020-78601-x>.
- Tian, H., Y. Wu, J. Tian, Y. Xu, Y. Cao, K. Xu, and Y. Feng. 2022. "The Effect of ZnO Particle Size on the Dynamic Mechanical, Thermal, and Dielectric Properties of ZnO Varistor-Epoxy Composites." *Journal of Materials Science: Materials in Electronics* 33 (28). Springer US:22388–22399. <https://doi.org/10.1007/s10854-022-09016-9>.
- Torres, M., V. Renteria Rodriguez, P. Itzel Alcantara, and E. Franco-Urquiza. 2022. "Mechanical Properties and Fracture Behaviour of Agave Fibers Bio-Based Epoxy Laminates Reinforced with Zinc Oxide." *Journal of Industrial Textiles* 51 (4): 5847S–5868S. <https://doi.org/10.1177/1528083720965689>.
- Turlybekuly, A., A. D. Pogrebnyak, L. F. Sukhodub, L. B. Sukhodub, A. S. Kistaubayeva, I. S. Savitskaya, D. H. Shokatayeva, et al. 2019. "Synthesis, Characterization, in vitro Biocompatibility and Antibacterial Properties Study of Nanocomposite Materials Based on Hydroxyapatite-Biphasic ZnO Micro-And Nanoparticles Embedded in Alginate Matrix." *Materials Science & Engineering: C* 104. Elsevier:109965. 109965. <https://doi.org/10.1016/j.msec.2019.109965>.
- Wong, T.-T., K.-T. Lau, W.-Y. Tam, J. A. Etches, J.-K. Kim, and Y. Wu. 2016. "Effects of Silane Surfactant on Nano-ZnO and Rheology Properties of Nano-ZnO/epoxy on the UV Absorbability of Nano-ZnO/epoxy/micron-HGF Composite." *Composites Part B Engineering* 90. Elsevier:378–385. <https://doi.org/10.1016/j.compositesb.2016.01.005>.
- Wong, T.-T., K.-T. Lau, W.-Y. Tam, J. Leng, and J. A. Etches. 2014. "UV Resistibility of a Nano-ZnO/glass Fibre Reinforced Epoxy Composite." *Materials & Design (1980-2015)* 56. Elsevier:254–257. <https://doi.org/10.1016/j.matdes.2013.11.014>.
- Wu, J., Q. Sun, H. Huang, Y. Duan, G. Xiao, and T. Le. 2019. "Enhanced Physico-Mechanical, Barrier and Antifungal Properties of Soy Protein Isolate Film by Incorporating Both Plant-Sourced Cinnamaldehyde and Facile Synthesized Zinc Oxide Nanosheets." *Colloids and Surfaces B, Biointerfaces* 180. Elsevier:31–38. <https://doi.org/10.1016/j.colsurfb.2019.04.041>.
- Yadav, Y. K., G. Dixit, and S. Dixit. 2023. "Natural Fiber Reinforced RPET/Polyester Composites: A Review on Development, Mechanical Performance, and Sustainable Management." *Polymer-Plastics Technology & Materials* (00). Taylor & Francis:1–21. <https://doi.org/10.1080/25740881.2023.2237100>.
- Yousry, M., M. Zaghloul, M. Mahmoud, Y. Zaghloul, M. Mahmoud, and Y. Zaghloul. 2021. "Developments in Polyester Composite Materials – an In-Depth Review on Natural Fibres and Nano Fillers." *Composite Structures* 278 (July). Elsevier Ltd. 114698. <https://doi.org/10.1016/j.compstruct.2021.114698>.
- Yu, H., G. Chen, Y. Wang, and J. Yao. 2015. "A Facile One-Pot Route for Preparing Cellulose Nanocrystal/Zinc Oxide Nanohybrids with High Antibacterial and Photocatalytic Activity." *Cellulose* 22 (1): 261–273. <https://doi.org/10.1007/s10570-014-0491-0>.
- Zongyu, W., M. R. Bockstaller, and K. Matyjaszewski. 2021. "Synthesis and Applications of ZnO/Polymer Nanohybrids." *ACS Materials Letters* 3 (5): 599–621. <https://doi.org/10.1021/acsmaterialslett.1c00145>.